

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY
(Deemed University)
M.Tech/MS/PhD Examinations
END SEMESTER - Examination (Spring, 2023)

Course Name: Machine Learning
Duration : 03 Hours

Course Code: AM 623/623D
Max Marks : 50

Note:

1. Marks are indicated against each question. Write clearly and legibly. All diagrams are to be neatly labelled. Calculators are allowed.
2. Attempt **ALL** questions.
3. Before attempting please ensure you have the correct question paper.

Q1	a)	Explain in detail what is needed for applying the machine learning algorithms to any given problem	2M										
	b)	With respect to Regression- explain the terms overfitting, underfitting, and the right fit of the model with diagrams. Explain any two techniques that can address the problem of underfitting.	5M										
	c)	What is feature normalization and why it is needed? How the Data: [10, 7, 15, 74, 99, 25] transforms after applying mean normalization	3M										
Q2	a)	Explain Polynomial regression in detail, and on what basis, Polynomial regression will be chosen over linear regression	5M										
	b)	Explain the Coefficient of Determination (R^2), and compute the R^2 value for Hypothesis: $h(x)=2x+3$, considering the below data.	5M										
		<table border="1"> <tr> <td>x (input)</td> <td>1</td> <td>2</td> <td>3</td> <td>5</td> </tr> <tr> <td>y (output)</td> <td>4.8</td> <td>7.2</td> <td>8.6</td> <td>12.8</td> </tr> </table>	x (input)	1	2	3	5	y (output)	4.8	7.2	8.6	12.8	
x (input)	1	2	3	5									
y (output)	4.8	7.2	8.6	12.8									
Q3	a)	Suppose you train a logistic regression classifier $h_\theta(x)=g(\theta_0 + \theta_1x_1 + \theta_2x_2 + \theta_3x_3)$, where g : sigmoid function, $\theta_0=-6$, $\theta_1=2$, $\theta_2=3$, $\theta_3=5$. By applying classifier $h_\theta(x)$ on the data points: A (1,1, 1), B(-1,1,-1), C(2,1, 1), D(4,-1,0), mention which data points belong to the same category.	5 M										
	b)	Explain in detail the working of the Hard margin SVM Classifier	5M										
Q4	a)	Draw the Decision Tree for the following Boolean functions	5 M										
	i)	$A \wedge \neg B$											
	ii)	$A \vee [B \wedge C]$											

	b	Give out any two reasons why classification based on Neural Network is preferred over logistic regression.	5 M
Q5	a)	Explain in detail how Naïve Bayes classifier works?	5 M
	b)	Apply K- means clustering algorithm with $K=2$ on the data [10, 25, 3, 10, 17, 12, 40, 64, 14, 52]. Use 25 and 40 as starting cluster centers. Mention separately the data points of two clusters and the cluster centroid. To which cluster the data <u>30</u> belongs to?	5 M

ALL THE BEST
